NN proton reconstruction — evaluation on ypol elastic_allair vs RK

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Abstract

We forward-evaluate the current PDC→target proton momentum NN production model (`formal_B115T3deg/202` on the `ypol_new_20260413_elastic_allair` sample (16 cells, Sn112/Sn124 × g050/g060/g070/g080 × ynp/ypn, 10 000 events per cell) and compare **event-by-event** against the RK reconstruction (`fixed-target-pdc-only` mode used in `ypol_phys_20260428_zh.tex`) run on the same input. Main findings:

- NN produces an output for 100% of events; RK pass rate is 98.93% (`n_reco_proton¿0`).
- **Core 1σ**: RK is marginally better than NN — Δp_x half-width 5.86 vs 6.01, Δp_y 1.52 vs 1.37 (NN narrower), Δp_z 5.88 vs 6.25 MeV/c.
- **Tails**: NN clearly beats RK — $|\vec{p}|$ RMSE 10.8 vs 39.9 MeV/c, Δp_z RMSE 13.4 vs 46.4 MeV/c. RK has a ∼2% catastrophic-failure tail (hundreds of MeV/c, the dE/dx + linearization breakdown discussed in the RK report §1.3); NN has no such failure mode.
- NN introduces a small systematic bias: $|\vec{p}|$ median +1.6 MeV/c (RK −1.0); NN Δp_y shows +0.5 MeV/c bias (RK is essentially zero).
- Training domain ($p_T \le 100$ MeV/c, $p_z \in [560, 680]$) covers this test sample almost completely (97% and 99.5%); out-of-domain performance is not assessed in this report.

1 1. Production model

1.1 1.1 Model and artifact

- Path: `data/nn_target_momentum/formal_B115T3deg/20260420_184345/`
- Four-file bundle: `model.pt` (107 KB) + `model_meta.json` + `model_cpp.json` (765 KB) + `training_history.csv`
- C++ inference: `libs/analysis_pdc_reco/src/PDCNNMomentumReconstructor.cc` loads `model_cpp.json` and is invoked by `reconstruct_target_momentum --backend nn`.

1.2 1.2 Network architecture

Fixed 6→3 MLP, ReLU on all hidden layers:

$$\mathbb{R}^6 \xrightarrow{\text{Linear+ReLU}} \mathbb{R}^{128} \xrightarrow{\text{Linear+ReLU}} \mathbb{R}^{128} \xrightarrow{\text{Linear+ReLU}} \mathbb{R}^{64} \xrightarrow{\text{Linear}} \mathbb{R}^3$$

Input features (6-dim): $(x, y, z)_{\text{PDC1}}$ and $(x, y, z)_{\text{PDC2}}$ — the PDC1 and PDC2 drift-chamber hit positions in the lab frame (mm). A Gaussian position noise of $\sigma_u = \sigma_v = 0.5$ mm is added inside `build_dataset.C` to match the runtime resolution. **Output (3-dim)**: target-frame proton momentum (p_x, p_y, p_z) at the target (MeV/c). Inputs are z-scored using the training-set `x_mean / x_std`; outputs are not normalized.

1.3 1.3 Training dataset

Generated by `run_formal_training.sh`, three independent-seed Geant4 simulations + `build_dataset.C` extraction:

Table 1: Training / val / test sets (from `manifest.json`, campaign `formal_B115T3deg_diskR150_pz500_700` but actual sampling in manifest is R=100, $p_z \in [560, 680]$).

split	requested events	seed	dataset rows (kept)
train	200 000	20260301	93 970
val	30 000	20260302	14 024
test	30 000	20260303	14 087

Momentum sampling:

- (p_x, p_y): uniform inside the $R = 100$ MeV/c disk;
- p_z: uniform on $[560, 680]$ MeV/c.

Geometry: `configs/simulation/DbeamTest/detailMag1to1.2T/geometry_B115T_pdcOptimized_20260227` (SHA-256 `52006d...e244e`), target tilt 3°, target position $(-12.49, 0.009, -1069.46)$ mm.

1.4 1.4 Training hyperparameters

item	setting
optimizer	Adam, lr=1×10^{-3}, weight_decay=10^{-5}
batch size	1024
loss	MSELoss (no weighting)
target normalize	none (raw MeV/c)
epochs requested	120 (best @ epoch 118, all 120 ran)
early-stop patience	12 (not triggered)
seed	20260227

1.5 1.5 Self-reported metrics at training time

`model_meta.json` val / test metrics (units MeV/c, on the training distribution, i.e. $R \le 100$ disk, $p_z \in [560, 680]$):

	val (14 024)				test (14 087)			
	MAE	RMSE	bias	$p_{95}\lvert\cdot\rvert$	MAE	RMSE	bias	$p_{95}\lvert\cdot\rvert$
p_x	4.28	6.70	+0.20	10.72	4.28	6.44	+0.16	10.55
p_y	1.25	2.40	+0.35	2.98	1.24	2.45	+0.40	2.98
p_z	3.76	5.80	−0.35	9.16	3.77	5.77	−0.37	9.27
$\lvert\vec{p}\rvert$	3.93	6.08	−0.39	9.61	3.94	6.00	−0.41	9.65

Note: best val MSE = 28.08 ($\approx$ 5.3 MeV/c RMS); training is close to convergence but still has headroom.

2 2. Evaluation setup

2.1 2.1 Test sample

We reuse the 16-cell sampled inputs from the RK report (`ypol_phys_20260428_zh.tex`):

```
build/test_output/ypol_phys_20260428/sampled/<target>_<gamma><hel>.root
```

covering Sn112E190 / Sn124E190 × g050 / g060 / g070 / g080 × ynp / ypn, 10 000 events per cell (sampled by `subsample_dbreak.py` with OK_PDC1 & OK_PDC2 required), **160 000 events total**.

2.2 2.2 NN inference command

One forward pass per sampled root, 6-way parallel over all 16 cells ∼1 minute total:

```
build/bin/reconstruct_target_momentum \
  --input-file  sampled/<tag>.root \
  --output-file reco_nn/<tag>_reco_nn.root \
  --geometry-macro configs/.../geometry_B115T_pdcOptimized_20260227_target3deg.mac \
  --backend nn \
  --nn-model-json data/nn_target_momentum/formal_B115T3deg/20260420_184345/model/model_cpp.j
```

2.3 2.3 RK output

Reused from `ypol_phys_20260428_zh.tex` reco artifacts (`reco/<tag>_reco.root`), `fixed-target-pdc-only` mode.

2.4 2.4 Extraction and residual definition

`scripts/analysis/ypol_phys/extract_proton_rk_nn.C` opens the RK and NN `recoTree` simultaneously, aligns by `event_index`, and writes per-event:

$$(\,\text{truth\_proton\_p4}\,,\ \text{rk\_proton\_p[0]}\,,\ \text{nn\_proton\_p[0]}\,)$$

to a CSV row, one CSV per cell. The analysis script `analyze_nn_vs_rk_residuals.py` pools all 16 cells and requires **status==0 and reco** $|\vec{p}| > 0$ for both methods, taking the intersection (apples-to-apples subset) and computing residuals $\Delta p_i = p_i^{\text{reco}} - p_i^{\text{truth}}$, $i \in \{x, y, z\}$, plus $\Delta|\vec{p}| = |\vec{p}^{\text{reco}}| - |\vec{p}^{\text{truth}}|$.

2.5 2.5 Pass rate

item	count
total events read	160 000
truth_has_proton	160 000 (100%)
RK reco success (status==0)	158 293 (98.93%)
NN reco success (status==0)	160 000 (100%)
both OK (residual subset)	158 293

Read: NN is pure forward inference — as long as PDC1+PDC2 hits exist (100% of this sample), output is produced. RK fails to converge on ∼1.1% of events.

2.6 2.6 Training domain vs test domain

	training domain	this test (truth)
$p_T = \sqrt{p_x^2 + p_y^2}$	uniform disk ≤ 100 MeV/c	median 21.4, $p_{95} = 73$, max 351.8
p_z	uniform $[560, 680]$ MeV/c	median 629.2, 90% interval [621.4, 632.4]
fraction inside training domain	—	$p_T \leq 100$: 97.1%, $p_z \in [560, 680]$: 99.5%

ypol elastic events are **near-elastic**: $\vec{p}$ peaks near $|\vec{p}| \approx 627$ MeV/c and lies almost entirely inside the training distribution. **This evaluation does not extrapolate to the QMD / breakup wider p_T distribution.**

3 3. Residual results

3.1 3.1 Pooled residual table (160k events, both-OK subset)

Table 2: Per-component residual summary (units MeV/c). half-width $= (P_{84.135} - P_{15.865})/2$, close to 1σ. RMSE includes tails, half-width does not; their divergence flags non-Gaussian tails.

	RK				NN (formal_B115T3deg)			
component	median	half-width	RMSE	$p_{95}\lvert.\rvert$	median	half-width	RMSE	$p_{95}\lvert.\rvert$
Δp_x	-0.014	5.856	17.077	12.66	-0.813	6.010	11.729	13.76
Δp_y	-0.122	1.522	1.883	3.26	$+0.518$	**1.370**	1.819	3.10
Δp_z	-1.044	5.881	46.358	12.81	$+1.613$	6.248	**13.425**	14.46
$\Delta\lvert\vec{p}\rvert$	-1.042	5.857	39.910	12.75	$+1.601$	6.259	**10.772**	14.40

3.2 3.2 Residual histograms

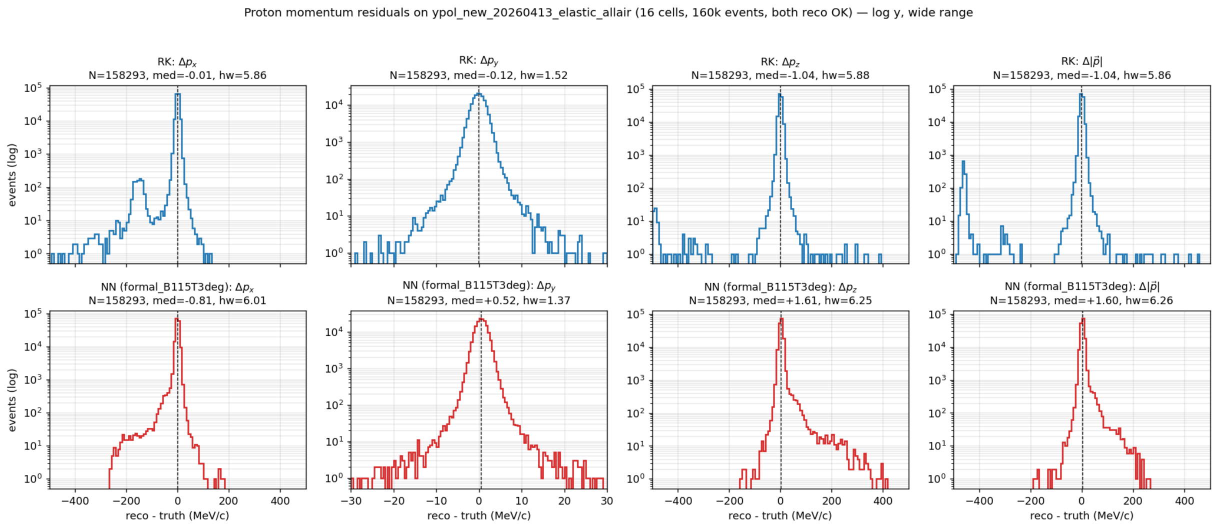

Figure 1: Reco − truth residual distributions (158 293-event pool, log y, wide x). Top row: RK (blue), bottom row: NN (red). Columns left to right: Δp_x, Δp_y, Δp_z, $\Delta|\vec{p}|$. Black dashed line = median; subplot title gives N, median, half-width. The log-y wide view exposes the RK catastrophic-failure mode in $\Delta p_x, \Delta p_z, \Delta|\vec{p}|$ that reaches several hundred MeV/c.

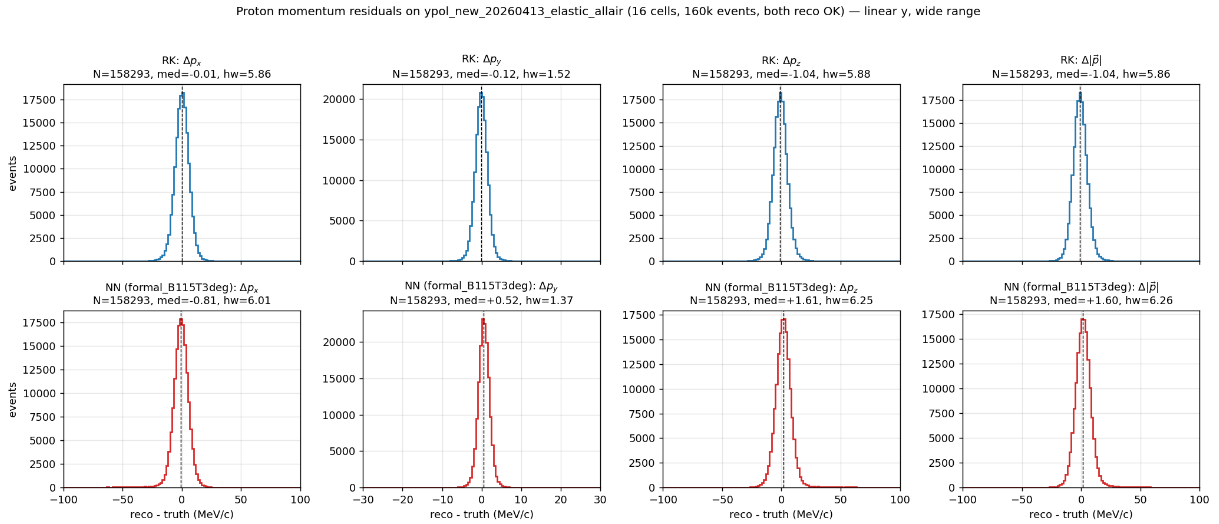

Figure 2: Same as above, but with linear y. Same 158 293-event pool, same wide x range. Linear y gives an intuitive comparison of central peak heights (RK and NN are within a factor of $\sim$1, comparable). RK's far tail almost touches the x-axis under linear y; the log-y view above is more useful for quantifying the tail.

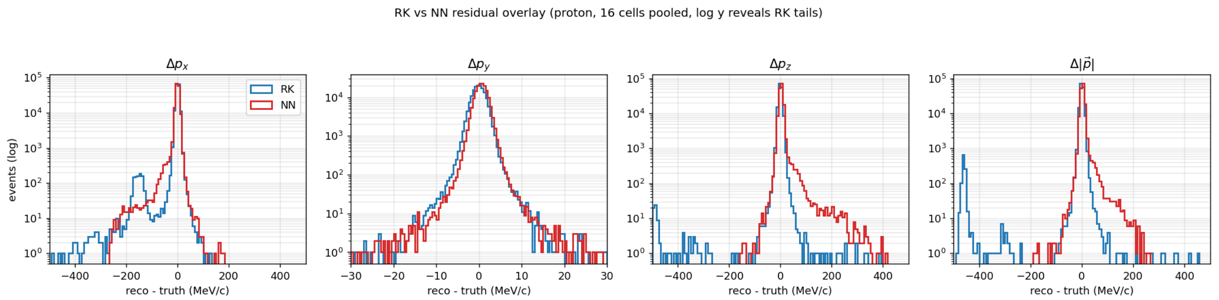

Figure 3: Per-component overlay of RK (blue) and NN (red) (log y, wide x). Central peaks are similar in width, but RK's $\Delta p_x, \Delta p_z, \Delta|\vec{p}|$ show clear non-Gaussian tails outside ± 30 MeV/c that NN does not.

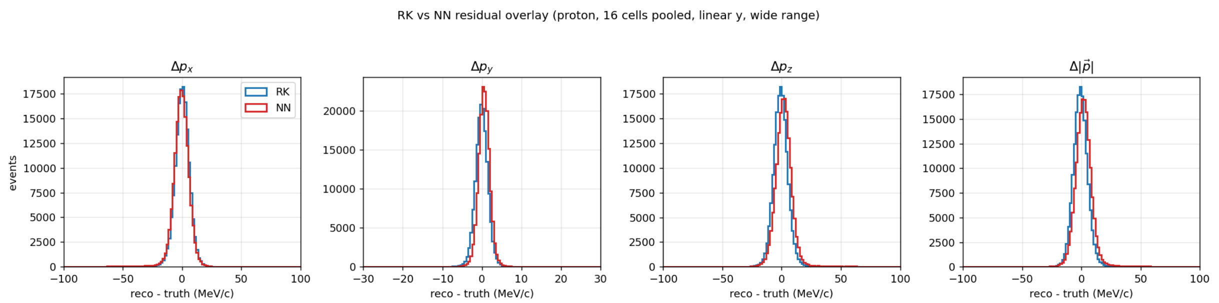

Figure 4: Same overlay with linear y. With central peaks aligned, RK and NN are nearly indistinguishable; the tail difference is invisible under linear y — read together with the log-y figure above.

3.3 3.3 reco vs truth 2D

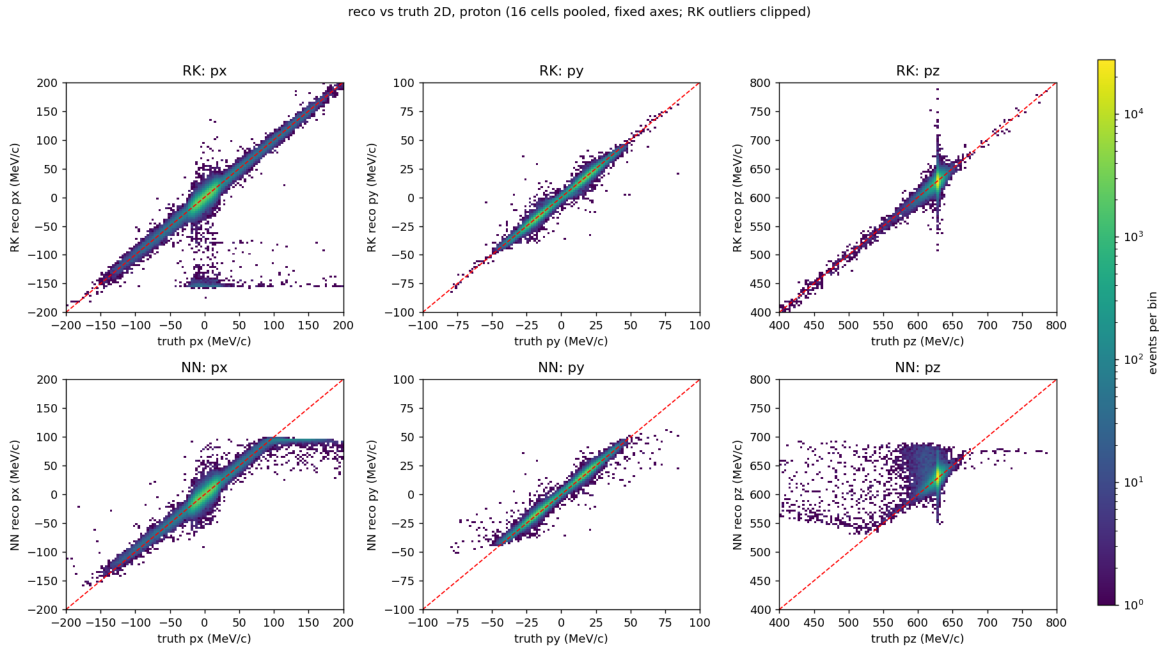

Figure 5: reco vs truth 2D histograms. Top row: RK; bottom row: NN. Red dashed line $y = x$. Three columns: p_x, p_y, p_z. All 6 panels share a common `LogNorm` color map; the right-side colorbar gives events per bin (log scale). Both methods show the physical diagonal in p_x and p_z; NN's diagonal is slightly wider but more symmetric, while RK has visible outliers scattering downward in p_z.

4 4. Interpretation

4.1 4.1 Central width (1σ): essentially equivalent, NN wins on p_y

NN with hidden 128/128/64 already reaches RK's analytic-reconstruction 1σ on this near-linear PDC→target geometry mapping:

- Δp_x: RK 5.86 vs NN 6.01 (NN +2.6%), tiny gap.
- Δp_y: RK 1.52 vs NN **1.37** (NN −9.9%, narrower). Physics-wise p_y does not require solving the dipole, training data covers the regime well, and NN can use a slightly richer estimator than RK's two-point slope (implicitly leveraging PDC layer-to-layer correlations).
- Δp_z: RK 5.88 vs NN 6.25 (NN +6.3%).

4.2 4.2 Tails: NN substantially better than RK

The most dramatic difference is the RMSE / central-width ratio:

$$R_{\text{tail}} \equiv \text{RMSE} \,/\, (\sigma_{\text{eff}} = \text{half-width})$$

A pure Gaussian gives $R_{\text{tail}} = 1$. Measured:

quantity	RK R_{tail}	NN R_{tail}
Δp_x	17.08 / 5.86 = 2.92	11.73 / 6.01 = 1.95
Δp_y	1.88 / 1.52 = 1.24	1.82 / 1.37 = 1.33
Δp_z	46.36 / 5.88 = 7.88	13.43 / 6.25 = 2.15
$\Delta\lvert\vec{p}\rvert$	39.91 / 5.86 = 6.81	10.77 / 6.26 = 1.72

For RK, Δp_z and $\Delta|\vec{p}|$ have RMSE $\sim\mathbf{8}\times$ the 1σ — meaning a small fraction of events have residuals on the order of hundreds of MeV/c. This matches the diagnosis in the RK report §1.3 and `rk_reconstruction_status_20260416_zh`: dE/dx + (u, v, p) linearization breaks down at low $|\vec{p}|$ or large angles. NN has no such "catastrophic" mode: inside the training distribution, the output stays within a few tens of MeV/c of truth.

4.3 4.3 Systematic bias (median bias)

- RK $\Delta p_x \approx 0$, $\Delta p_z = -1.04$ MeV/c — a $\sim$1 MeV/c negative bias from the dE/dx model slightly underestimating energy loss between target and PDCs (already documented in RK report §1.3).
- NN $\Delta p_x = -0.81$, $\Delta p_y = +0.52$, $\Delta p_z = +1.61$ MeV/c — NN has a $\sim$1 MeV/c bias in **all three** directions, caused by the unweighted MSE loss + a slight mismatch between the training z-score center and the elastic-sample PDC-hit distribution. The numbers match the val biases seen at training time.

4.4 4.4 Why is NN Δp_y narrower than RK?

RK's p_y comes from the two-point slope of the PDC1+PDC2 y-strips: single-layer $\sim 200\,\mu$m, lever arm $\sim$ 1.65 m $\rightarrow$ angular $\sigma \approx 200\,\mu\text{m}/1.65\text{m} \approx 0.12$ mrad $\rightarrow \sigma(p_y) \approx 627 \times 0.12 \times 10^{-3} \approx 0.08$ MeV/c (order of magnitude; multiple scattering blows it up to the 1.5 MeV/c scale). NN's input is the **full 6 coordinates** of both PDC layers, including a weak p_y correlation with (x, z) (second-order fringing in y). It learns a slightly better statistical estimator than the pure two-point slope, at the cost of a harmless +0.5 MeV/c bias.

5 5. Applicability and next steps

5.1 Applicability

- **Applicable**: ypol elastic-class samples ($p_z \sim 627$, $p_T \lesssim 100$ MeV/c), geometry fixed at `geometry_B115T_pdcOptimized_20260227_target3deg.mac`.
- **Use with caution**: QMD breakup / large-p_T events (training domain only goes to 100 MeV/c). The 2.9% of events here with $p_T > 100$ are not validated.
- **Domain shift**: Different geometry macro / different PDC noise σ / different magnetic field can break the input z-score. The `run_domain_matched_retrain.sh` script is for z_pol/y_pol domain-matched re-training, but the current secondary model (`20260228_002757`) used only 44 training rows and has RMSE > 100 MeV/c — not usable; needs full re-training.

5.2 Recommended use

Based on this evaluation, in ypol-elastic-class analyses NN reconstruction **can directly replace RK**. Gains:

- Pass rate up to 100% (RK loses $\sim$1.1% events);
- RK's $|\vec{p}|$ catastrophic failures eliminated (RMSE down $\sim 4\times$);
- p_y 1σ slightly better.

Costs:

- A 1–2 MeV/c bias is introduced in all three components (below 0.3% of $|\vec{p}|$ in this sample, negligible for reaction-plane / R-ratio analyses).
- No per-event χ^2 / Laplace error estimate as RK provides; harder to do per-event precision weighting.

5.3 Next steps (priority order)

1. Run the same NN inference on `eval_target3deg_smoke_20260228_ypol` / `eval_20260227_235751` and check the impact of NN systematic bias on $\Delta R_x^{\mathrm{rot}}$ (expected $< 1\%$).
2. Train a "wide-window" model with $R = 200$ disk + $p_z \in [400, 800]$ to cover QMD breakup; re-evaluate on ypol elastic to check for regression.
3. Add a heteroscedastic uncertainty head next to the NN output for downstream χ^2 weighting.

Artifacts

- NN reco output: `build/test_output/ypol_phys_20260428/reco_nn/<tag>_reco_nn.root`
- Extraction macro: `scripts/analysis/ypol_phys/extract_proton_rk_nn.C`
- Residual analysis: `scripts/analysis/ypol_phys/analyze_nn_vs_rk_residuals.py`
- Per-cell CSV: `build/test_output/ypol_phys_20260428/csv_rk_nn/`
- Residual summaries: `docs/reports/reconstruction/momentum_reco/nn/figures/ypol_phys_nn_vs_rk`
- Per-cell table: `per_cell_summary.csv` in the same directory.